

1. Questions

Study the following information carefully and answer the questions given below.

In three different schools A, B and C the total number of students enrolled is 1200. The ratio of the total number of students in School A, School B and School C is 7 : 8 : 9. In School A, the number of boys is 50 more than the number of girls. In School B, the ratio of the number of boys to girls is 3 : 2. In School C, the number of girls is 25% more than the number of boys.

Find the difference between the total number of girls in School A and School B together and the total number of boys in School B and School C together.

- a. 140
- b. 110
- c. 120
- d. 130
- e. 150

2. Questions

If 25% of the boys in School B and 30% of the girls in School C participate in sports, find the total number of students who do not participate in sports in their respective Schools.

- a. 255
- b. 355
- c. 155
- d. 455
- e. 555

3. Questions

What is the ratio of the number of girls in School A to the number of boys in School C?

- a. 5 : 6
- b. 1 : 2
- c. 4 : 5
- d. 2 : 3
- e. 3 : 4

4. Questions

The number of girls in School B is what percent of the total number of students in School C?

- a. 33.33%

- b. 35.55%
- c. 45.45%
- d. 22.22%
- e. 19.19%

5. Questions

Find the total number of boys in all three schools together.

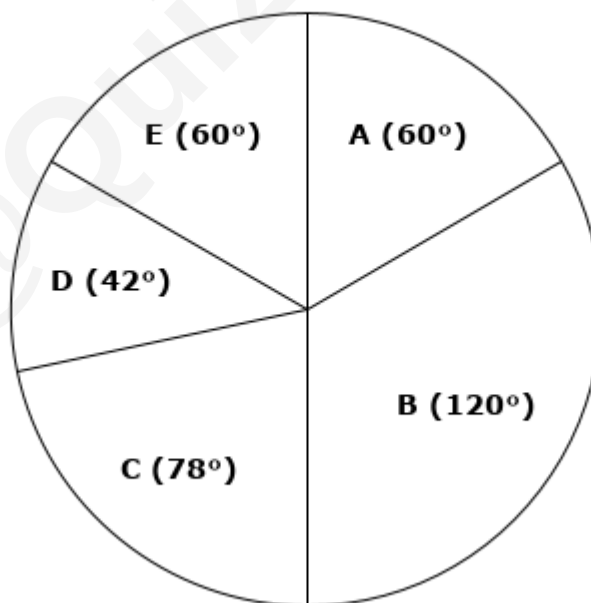
- a. 640
- b. 520
- c. 450
- d. 720
- e. 390

6. Questions

Study the following information carefully and answer the questions given below.

The pie chart given below shows the degree distribution of the number of TVs sold by five different shops namely A, B, C, D and E.

The degree distribution of TVs sold



Note: The total number of TVs sold by all shops together is 720.

If Shop F sold 25% more TVs than Shop C, and the ratio of defective to non-defective TVs sold by Shop F is 5 : 8, find the number of non-defective TVs sold by Shop F.

- a. 130

- b. 65
- c. 110
- d. 120
- e. 95

7. Questions

If the ratio between the number of LG brand TVs and ONIDA brand TVs sold by shop A is 7 : 5, respectively, find the number of LG brand TVs sold by shop A.

- a. 70
- b. 50
- c. 60
- d. 80
- e. 90

8. Questions

The number of TVs sold by shop D is what percentage less than the number of TVs sold by shop E?

- a. 20% less
- b. 30% more
- c. 15% less
- d. 20% more
- e. 30% less

9. Questions

Find the ratio of the difference of the number of TVs sold by shops B and E to the number of TVs sold by shop C.

- a. 9 : 8
- b. 10 : 13
- c. 11 : 8
- d. 13 : 17
- e. 6 : 7

10. Questions

The number of TVs sold by shop C is how much more or less than the number of TVs sold by shop D?

- a. 60
- b. 36
- c. 72
- d. 84
- e. 96

11. Questions

The length of a rectangle is 62.5% of the side of a square whose area is 576 cm^2 . If the breadth of the rectangle is 7 cm less than its length, find the area of the rectangle.

- a. 144 cm^2
- b. 120 cm^2
- c. 135 cm^2
- d. 150 cm^2
- e. 128 cm^2

12. Questions

In college X, the ratio of the number of boys to girls is 5 : 4. On a particular day, 20% of the boys and 25% of the girls were absent. If the total number of students present on that day was 210, find the total number of girls enrolled in the college.

- a. 120
- b. 180
- c. 140
- d. 150
- e. 190

13. Questions

If a sum is compounded annually at a rate of $16 \frac{2}{3}\%$ for two years and the compound interest for the 2nd year is Rs. 84, find the sum.

- a. Rs. 549
- b. Rs. 658
- c. Rs. 543
- d. Rs. 420
- e. Rs. 432

14. Questions

Ram started a business with an investment of Rs. 1500. After 6 months, Shyam joined him with an investment of Rs. x . If Shyam left the business 10 months after it started and his share of the total annual profit was 33.33%, find the value of x .

- a. Rs. 2150
- b. Rs. 2240
- c. Rs. 2130
- d. Rs. 2250
- e. Rs. 2430

15. Questions

The sum of the squares of two positive numbers is 954, and one number is $44\frac{4}{9}\%$ less than the other number. Find the larger number.

- a. 18
- b. 27
- c. 24
- d. 15
- e. 12

16. Questions

M, N, and P can complete a piece of work in 20, 24 and 30 days respectively. M and N start the work together. After 3 days, M leaves the work and P joins with N. They work together for another 3 days, after which N leaves. In how many days will P complete the remaining work alone?

- a. 10 days
- b. 12 days
- c. 15 days
- d. 18 days
- e. 9 days

17. Questions

Train A of length x meters and Train B of length $(x + 40)$ meters are running in opposite directions on parallel tracks. The speed of Train A is 90 km/hr and the speed of Train B is 54 km/hr. If they cross each other completely in 11 seconds, find the time taken by Train A to cross the platform of length $(x + 150)$ meters.

- a. 22 seconds

- b. 18 seconds
- c. 33 seconds
- d. 15 seconds
- e. 11 seconds

18. Questions

The ratio of the present ages of P and R is 5 : 4, respectively. The present age of S is 50% more than the present age of R. If the ratio of the age of P to that of S, 6 years hence will be 6 : 7, find the present age of P.

- a. 35 years
- b. 24 years
- c. 32 years
- d. 28 years
- e. 30 years

19. Questions

A vessel contains 300 litres of a milk to water mixture in the ratio 3 : 2. 60 litres of the mixture is drawn out and replaced with water. Thereafter, 50 litres of the resulting mixture is again drawn out and replaced with water. Find the quantity of milk left in the vessel.

- a. 150 litres
- b. 130 litres
- c. 140 litres
- d. 120 litres
- e. 160 litres

20. Questions

The cost prices of two articles are in the ratio 5 : 7, respectively. A shopkeeper marked up the first article by 40% above its cost price, offered a discount of 15% on it, and sold it for Rs. 2380. Find the selling price of the second article if it is sold at a profit of 25%.

- a. Rs. 3200
- b. Rs. 3500
- c. Rs. 3100
- d. Rs. 3600
- e. Rs. 3300

21. Questions

What value should come in the place of (?) in the following questions.

$$45.5 + 14.5 * 4 - 28 = ? + 18.5 * 2$$

- a. 41.5
- b. 38.5
- c. 42.5
- d. 37.5
- e. 43.5

22. Questions

$$(8.5 * 16) + (15 * 18) + (12 * 19) = ?$$

- a. 634
- b. 523
- c. 628
- d. 596
- e. 655

23. Questions

$$(12.5 + 6.5)^2 + 14 * 7 - ? = 320$$

- a. 142
- b. 123
- c. 139
- d. 138
- e. 127

24. Questions

$$324 \div 3^2 - 24 + 31 = ?$$

- a. 48
- b. 27
- c. 41
- d. 39
- e. 43

25. Questions

$$[720 \div 2^2 \div 3] * ? = 300 + 40 * 6$$

- a. 3
- b. 7
- c. 4
- d. 9
- e. 5

26. Questions

What approximate value will come in the place of (?) in the following questions?

$$45.02\% \text{ of } 1599.89 + \sqrt{728.87} = 349.98 + ?$$

- a. 397
- b. 323
- c. 368
- d. 379
- e. 352

27. Questions

$${}^3\sqrt{63999.98} - \sqrt{624.89} - {}^3\sqrt{343.30} = ?^2 * 2.2 - 23.98$$

- a. 7
- b. 3
- c. 4
- d. 2
- e. 9

28. Questions

$$(72.20 \div 7.89) * (83.98 \div 7.01) + ? = 44.98\% \text{ of } 299.97$$

- a. 38
- b. 27
- c. 19
- d. 42
- e. 23

29. Questions

$$(13.98 * 15.02) \div 5.94 + 11.01^2 = ? * 3.98$$

- a. 31
- b. 28
- c. 41
- d. 39
- e. 17

30. Questions

$$44.99\% \text{ of } 1399.98 + \sqrt{576.23} = 499.87 + ?$$

- a. 176
- b. 167
- c. 134
- d. 145
- e. 154

31. Questions

What value will come in the place of (?) in the following number series?

83, 97, 125, 167, 223, ?

- a. 254
- b. 293
- c. 268
- d. 275
- e. 247

32. Questions

1600, 1650, 1625, 1675, ?, 1700

- a. 1650
- b. 1550
- c. 1450
- d. 1750
- e. 1350

33. Questions**90, 91, 94, ?, 130, 211**

- a. 110
- b. 124
- c. 103
- d. 132
- e. 106

34. Questions**15, 17, 22, 32, 49, ?**

- a. 53
- b. 61
- c. 72
- d. 75
- e. 59

35. Questions**389, 289, ?, 144, 95, 59**

- a. 206
- b. 212
- c. 204
- d. 210
- e. 208

36. Questions**Find the wrong number in the following number series.****131, 139, 130, 194, 169, 203**

- a. 131
- b. 203
- c. 130
- d. 194
- e. 169

37. Questions**43, 49, 55, 66, 81, 103**

- a. 55
- b. 43
- c. 49
- d. 66
- e. 81

38. Questions**120, 128, 146, 182, 254, 398**

- a. 398
- b. 182
- c. 146
- d. 128
- e. 120

39. Questions**249, 272, 301, 332, 369, 411**

- a. 332
- b. 301
- c. 249
- d. 411
- e. 369

40. Questions**41, 122, 222, 343, 485, 656**

- a. 122
- b. 41
- c. 485
- d. 656
- e. 343

41. Questions

The following questions contain two equations I and II. Solve both equations and determine the relationship between them and give answer as,

I). $x^2 - 23x + 132 = 0$

II). $y^2 - 19y + 88 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

42. Questions

I). $x^2 - 17x + 72 = 0$

II). $y^2 - 9y + 18 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

43. Questions

I). $x^2 - 27x + 170 = 0$

II). $y^2 - 36y + 323 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

44. Questions

I). $x^2 - 8x + 15 = 0$

II). $y^2 - 13y + 42 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

45. Questions

I). $x^2 - 49 = 0$

II). $y^2 - 12y + 35 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

Explanations:

1. Questions

The total number of students in all the schools together = 1200

The number of students in school A = $1200 * \frac{7}{24} = 350$

The number of students in school B = $1200 * \frac{8}{24} = 400$

The number of students in school C = $1200 * \frac{9}{24} = 450$

The number of boys in school A = $(350 + 50)/2 = 200$

The number of girls in school A = $(350 - 50)/2 = 150$

Let the number of boys in school B = $3a$

The number of girls in school B = $2a$

So,

$$3a + 2a = 400$$

$$5a = 400$$

$$a = 80$$

The number of boys in school B = $3 * 80 = 240$

The number of girls in school B = $2 * 80 = 160$

Let the number of boys in school C = $4b$

The number of girls in school C = $4b * 125/100 = 5b$

So,

$$4b + 5b = 450$$

$$9b = 450$$

$$b = 50$$

The number of boys in school C = $4 * 50 = 200$

The number of girls in school C = $5 * 50 = 250$

Schools	The number of boys	The number of girls	The total number of students together
A	200	150	350
B	240	160	400
C	200	250	450
Total	640	560	1200

Answer: D

The total number of girls in schools A and B together = $150 + 160 = 310$

The total number of boys in schools B and C together = $240 + 200 = 440$

The required difference = $440 - 310 = 130$

Hence, the correct answer is option D.

2. Questions

The total number of students in all the schools together = 1200

The number of students in school A = $1200 * 7/24 = 350$

The number of students in school B = $1200 * 8/24 = 400$

The number of students in school C = $1200 * 9/24 = 450$

The number of boys in school A = $(350 + 50)/2 = 200$

The number of girls in school A = $(350 - 50)/2 = 150$

Let the number of boys in school B = $3a$

The number of girls in school B = $2a$

So,

$$3a + 2a = 400$$

$$5a = 400$$

$$a = 80$$

$$\text{The number of boys in school B} = 3 * 80 = 240$$

$$\text{The number of girls in school B} = 2 * 80 = 160$$

$$\text{Let the number of boys in school C} = 4b$$

$$\text{The number of girls in school C} = 4b * 125/100 = 5b$$

So,

$$4b + 5b = 450$$

$$9b = 450$$

$$b = 50$$

$$\text{The number of boys in school C} = 4 * 50 = 200$$

$$\text{The number of girls in school C} = 5 * 50 = 250$$

Schools	The number of boys	The number of girls	The total number of students together
A	200	150	350
B	240	160	400
C	200	250	450
Total	640	560	1200

Answer: B

$$\text{The number of boys in school B who do not participate in sports} = 240 * 75\% = 180$$

$$\text{The number of girls in school C who do not participate in sports} = 250 * 70\% = 175$$

$$\text{The required sum} = 180 + 175 = 355$$

Hence, the correct answer is option B.

3. Questions

$$\text{The total number of students in all the schools together} = 1200$$

$$\text{The number of students in school A} = 1200 * 7/24 = 350$$

$$\text{The number of students in school B} = 1200 * 8/24 = 400$$

$$\text{The number of students in school C} = 1200 * 9/24 = 450$$

The number of boys in school A = $(350 + 50)/2 = 200$

The number of girls in school A = $(350 - 50)/2 = 150$

Let the number of boys in school B = $3a$

The number of girls in school B = $2a$

So,

$$3a + 2a = 400$$

$$5a = 400$$

$$a = 80$$

The number of boys in school B = $3 * 80 = 240$

The number of girls in school B = $2 * 80 = 160$

Let the number of boys in school C = $4b$

The number of girls in school C = $4b * 125/100 = 5b$

So,

$$4b + 5b = 450$$

$$9b = 450$$

$$b = 50$$

The number of boys in school C = $4 * 50 = 200$

The number of girls in school C = $5 * 50 = 250$

Schools	The number of boys	The number of girls	The total number of students together
A	200	150	350
B	240	160	400
C	200	250	450
Total	640	560	1200

Answer: E

The number of girls in school A = 150

The number of boys in school C = 200

The required ratio = $150 : 200 = 3 : 4$

Hence, the correct answer is option E.

4. Questions

The total number of students in all the schools together = 1200

The number of students in school A = $1200 * \frac{7}{24} = 350$

The number of students in school B = $1200 * \frac{8}{24} = 400$

The number of students in school C = $1200 * \frac{9}{24} = 450$

The number of boys in school A = $(350 + 50)/2 = 200$

The number of girls in school A = $(350 - 50)/2 = 150$

Let the number of boys in school B = $3a$

The number of girls in school B = $2a$

So,

$$3a + 2a = 400$$

$$5a = 400$$

$$a = 80$$

The number of boys in school B = $3 * 80 = 240$

The number of girls in school B = $2 * 80 = 160$

Let the number of boys in school C = $4b$

The number of girls in school C = $4b * \frac{125}{100} = 5b$

So,

$$4b + 5b = 450$$

$$9b = 450$$

$$b = 50$$

The number of boys in school C = $4 * 50 = 200$

The number of girls in school C = $5 * 50 = 250$

Schools	The number of boys	The number of girls	The total number of students together
A	200	150	350
B	240	160	400
C	200	250	450
Total	640	560	1200

Answer: B

The number of girls in school B = 160

The total number of students in school C = 450

The required answer = $160 / 450 * 100 = 35.55\%$

Hence, the correct answer is option B.

5. Questions

The total number of students in all the schools together = 1200

The number of students in school A = $1200 * 7/24 = 350$

The number of students in school B = $1200 * 8/24 = 400$

The number of students in school C = $1200 * 9/24 = 450$

The number of boys in school A = $(350 + 50)/2 = 200$

The number of girls in school A = $(350 - 50)/2 = 150$

Let the number of boys in school B = $3a$

The number of girls in school B = $2a$

So,

$$3a + 2a = 400$$

$$5a = 400$$

$$a = 80$$

The number of boys in school B = $3 * 80 = 240$

The number of girls in school B = $2 * 80 = 160$

Let the number of boys in school C = $4b$

The number of girls in school C = $4b * 125/100 = 5b$

So,

$$4b + 5b = 450$$

$$9b = 450$$

$$b = 50$$

The number of boys in school C = $4 * 50 = 200$

The number of girls in school C = $5 * 50 = 250$

Schools	The number of boys	The number of girls	The total number of students together
A	200	150	350
B	240	160	400
C	200	250	450
Total	640	560	1200

Answer: A

The total number of boys in all three schools together = $200 + 240 + 200 = 640$

Hence, the correct answer is option A.

6. Questions

The total number of TVs sold by all shops together = 720

So,

$$360^\circ = 720$$

$$1^\circ = 2$$

The number of TVs sold by shop A = $60 * 2 = 120$

The number of TVs sold by shop B = $120 * 2 = 240$

The number of TVs sold by shop C = $78 * 2 = 156$

The number of TVs sold by shop D = $42 * 2 = 84$

The number of TVs sold by shop E = $60 * 2 = 120$

So,

Shops	The number of TVs sold by each shop
A	120
B	240
C	156
D	84
E	120

Answer: D

The number of TVs sold by shop F = $156 * \frac{5}{4} = 195$

The number of non-defective TVs sold by shop F = $195 * \frac{8}{13} = 120$

Hence, the correct answer is option D.

7. Questions

The total number of TVs sold by all shops together = 720

So,

$$360^\circ = 720$$

$$1^\circ = 2$$

The number of TVs sold by shop A = $60 * 2 = 120$

The number of TVs sold by shop B = $120 * 2 = 240$

The number of TVs sold by shop C = $78 * 2 = 156$

The number of TVs sold by shop D = $42 * 2 = 84$

The number of TVs sold by shop E = $60 * 2 = 120$

So,

Shops	The number of TVs sold by each shop
A	120
B	240
C	156
D	84
E	120

Answer: A

The total number of TVs sold by shop A = 120

The number of LG brand TVs sold by shop A = $120 * \frac{7}{12} = 70$

Hence, the correct answer is option A.

8. Questions

The total number of TVs sold by all shops together = 720

So,

$$360^\circ = 720$$

$$1^\circ = 2$$

The number of TVs sold by shop A = $60 * 2 = 120$

The number of TVs sold by shop B = $120 * 2 = 240$

The number of TVs sold by shop C = $78 * 2 = 156$

The number of TVs sold by shop D = $42 * 2 = 84$

The number of TVs sold by shop E = $60 * 2 = 120$

So,

Shops	The number of TVs sold by each shop
A	120
B	240
C	156
D	84
E	120

Answer: E

The number of TVs sold by shop D = 84

The number of TVs sold by shop E = 120

The required percentage = $(120 - 84) / 120 * 100 = 30\%$ less

Hence, the correct answer is option E.

9. Questions

The total number of TVs sold by all shops together = 720

So,

$$360^\circ = 720$$

$$1^\circ = 2$$

The number of TVs sold by shop A = $60 * 2 = 120$

The number of TVs sold by shop B = $120 * 2 = 240$

The number of TVs sold by shop C = $78 * 2 = 156$

The number of TVs sold by shop D = $42 * 2 = 84$

The number of TVs sold by shop E = $60 * 2 = 120$

So,

Shops	The number of TVs sold by each shop
A	120
B	240
C	156
D	84
E	120

Answer: B

The difference of the number of TVs sold by shops B and E = $240 - 120 = 120$

The number of TVs sold by shop C = 156

The required ratio = $120 : 156 = 10 : 13$

Hence, the correct answer is option B.

10. Questions

The total number of TVs sold by all shops together = 720

So,

$$360^\circ = 720$$

$$1^\circ = 2$$

The number of TVs sold by shop A = $60 * 2 = 120$

The number of TVs sold by shop B = $120 * 2 = 240$

The number of TVs sold by shop C = $78 * 2 = 156$

The number of TVs sold by shop D = $42 * 2 = 84$

The number of TVs sold by shop E = $60 * 2 = 120$

So,

Shops	The number of TVs sold by each shop
A	120
B	240
C	156
D	84
E	120

Answer: C

The number of TVs sold by shop C = 156

The number of TVs sold by shop D = 84

The required difference = $156 - 84 = 72$

Hence, the correct answer is option C.

11. Questions

Answer: B

The area of the square = a^2

So,

$$a^2 = 576$$

$$a = 24 \text{ cm}$$

The length of the rectangle = $24 * \frac{5}{8} = 15 \text{ cm}$

The breadth of the rectangle = $15 - 7 = 8 \text{ cm}$

The area of the rectangle = $15 * 8 = 120 \text{ cm}^2$

Hence, the correct answer is option B.

12. Questions

Answer: A

Let the number of boys = $5a$

The number of girls = $4a$

The number of present boys = $5a * \frac{4}{5} = 4a$

The number of present girls = $4a * \frac{3}{4} = 3a$

So,

$$4a + 3a = 210$$

$$7a = 210$$

$$a = 30$$

The total number of girls = $4 * 30 = 120$

Hence, the correct answer is option A.

13. Questions

Answer: E

$$[16(2/3)\% = 1/6]$$

Let the principal = $36a$

The compound interest for the first year = $36a * 1/6 = 6a$

The compound interest for the second year = $(36a * 1/6) + (6a * 1/6) = 6a + a = 7a$

According to the question,

$$7a = 84$$

$$a = 12$$

The invested sum = $36 * 12 = \text{Rs. } 432$

Hence, the correct answer is option E.

14. Questions

Answer: D

Let the total investment of Ram = $1500 * 12 = \text{Rs. } 18000$

The investment period of Shyam = $10 - 6 = 4$ months

The total investment of Shyam = $x * 4 = \text{Rs. } 4x$

According to the question,

$$18000/4x = 2/1$$

$$4x = 9000$$

$$x = 2250$$

Hence, the correct answer is option D.

15. Questions

Answer: B

Let the larger number = $9a$

The smaller number = $9a * 5/9 = 5a$

According to the question,

$$(9a)^2 + (5a)^2 = 954$$

$$81a^2 + 25a^2 = 954$$

$$106a^2 = 954$$

$$a^2 = 9$$

$$a = 3$$

The larger number = $9 * 3 = 27$

Hence, the correct answer is option B.

16. Questions

Answer: C

Let the total work = 120 units [LCM of 20, 24 and 30]

The efficiency of M = $120/20 = 6$ units per day

The efficiency of N = $120/24 = 5$ units per day

The efficiency of P = $120/30 = 4$ units per day

The work done by M and N in 3 days = $(6 + 5) * 3 = 33$ units

The work done by N and P in 3 days = $(5 + 4) * 3 = 27$ units

The remaining work = $120 - 33 - 27 = 60$ units

Then, P completes the remaining work alone = $60/4 = 15$ days

Hence, the correct answer is option C.

17. Questions

Answer: A

The speed of the train A = $90 * 5/18 = 25$ m/sec

The speed of the train B = $54 * 5/18 = 15$ m/sec

The relative speed of the train A and B = $25 + 15 = 40$ m/sec

According to the question,

$$(x + x + 40) / 11 = 40$$

$$2x = 440 - 40$$

$$2x = 400$$

$$x = 200$$

The length of the train A = 200

The length of the platform = $200 + 150 = 350$

The total distance travelled by train A = $200 + 350 = 550$ meters

The time taken by Train A to cross the platform = $550/25 = 22$ seconds

Hence, the correct answer is option A.

18. Questions

Answer: E

Let the present age of P = $5a$

The present age of R = $4a$

The present age of S = $4a * 150\% = 6a$

According to the question,

$$(5a + 6) / (6a + 6) = 6 / 7$$

$$35a + 42 = 36a + 36$$

$$a = 6$$

The present age of P = $5 * 6 = 30$ years

Hence, the correct answer is option E.

19. Questions

Answer: D

The initial quantity of the milk = $300 * 3/5 = 180$ litres

The initial quantity of the water = $300 * 2/5 = 120$ litres

The removed quantity of the milk = $60 * 3/5 = 36$ litres

The removed quantity of the water = $60 * 2/5 = 24$ litres

The remaining quantity of the milk = $180 - 36 = 144$ litres

The new quantity of the water = $120 - 24 + 60 = 156$ litres

The new ratio of milk and water = $144 : 156 = 12 : 13$

The removed quantity of the milk = $50 * 12/25 = 24$ litres

The removed quantity of the water = $50 * 13/25 = 26$ litres

The remaining quantity of the milk = $144 - 24 = 120$ litres

Hence, the correct answer is option D.

20. Questions

Answer: B

Let the cost price of the first article = $500a$

The cost price of the second article = $700a$

The marked price of the first article = $500a * 140/100 = 700a$

The selling price of the first article = $700a * 85\% = 595a$

According to the question,

$$595a = 2380$$

$$a = 4$$

The cost price of the second article = $700 * 4 = \text{Rs. } 2800$

The selling price of the second article = $2800 * 125/100 = \text{Rs. } 3500$

Hence, the correct answer is option B.

21. Questions

Answer: B

$$45.5 + 14.5 * 4 - 28 = ? + 18.5 * 2$$

$$45.5 + 58 - 28 = ? + 37$$

$$45.5 + 30 = ? + 37$$

$$75.5 = ? + 37$$

$$? = 38.5$$

Hence, the correct answer is option B.

22. Questions

Answer: A

$$(8.5 * 16) + (15 * 18) + (12 * 19) = ?$$

$$136 + 270 + 228 = ?$$

$$634 = ?$$

Hence, the correct answer is option A.

23. Questions

Answer: C

$$(12.5 + 6.5)^2 + 14 * 7 - ? = 320$$

$$19^2 + 98 - ? = 320$$

$$361 + 98 - ? = 320$$

$$459 - ? = 320$$

$$139 = ?$$

Hence, the correct answer is option C.

24. Questions**Answer: E**

$$324 \div 3^2 - 24 + 31 = ?$$

$$324 \div 9 - 24 + 31 = ?$$

$$36 - 24 + 31 = ?$$

$$12 + 31 = ?$$

$$43 = ?$$

Hence, the correct answer is option E.

25. Questions**Answer: D**

$$[720 \div 2^2 \div 3] * ? = 300 + 40 * 6$$

$$[720 \div 12] * ? = 300 + 240$$

$$60 * ? = 540$$

$$? = 9$$

Hence, the correct answer is option D.

26. Questions**Answer: A**

$$45.02\% \text{ of } 1599.89 + \sqrt{728.87} = 349.98 + ?$$

$$45\% \text{ of } 1600 + \sqrt{729} = 350 + ?$$

$$720 + 27 = 350 + ?$$

$$747 = 350 + ?$$

$$397 = ?$$

Hence, the correct answer is option A.

27. Questions**Answer: C**

$$^3\sqrt{63999.98} - \sqrt{624.89} - ^3\sqrt{343.30} = ?^2 * 2.2 - 23.98$$

$$^3\sqrt{64000} - \sqrt{625} - ^3\sqrt{343} = ?^2 * 2 - 24$$

$$40 - 25 - 7 = ?^2 * 2 - 24$$

$$8 = ?^2 * 2 - 24$$

$$32 = ?^2 * 2$$

$$16 = ?^2$$

$$? = 4$$

Hence, the correct answer is option C.

28. Questions

Answer: B

$$(72.20 \div 7.89) * (83.98 \div 7.01) + ? = 44.98\% \text{ of } 299.97$$

$$(72 \div 8) * (84 \div 7) + ? = 45\% \text{ of } 300$$

$$9 * 12 + ? = 135$$

$$108 + ? = 135$$

$$? = 27$$

Hence, the correct answer is option B.

29. Questions

Answer: D

$$(13.98 * 15.02) \div 5.94 + 11.01^2 = ? * 3.98$$

$$(14 * 15) \div 6 + 11^2 = ? * 4$$

$$35 + 121 = ? * 4$$

$$156 = ? * 4$$

$$39 = ?$$

Hence, the correct answer is option D.

30. Questions

Answer: E

$$44.99\% \text{ of } 1399.98 + \sqrt{576.23} = 499.87 + ?$$

$$45\% \text{ of } 1400 + \sqrt{576} = 500 + ?$$

$$630 + 24 = 500 + ?$$

$$654 = 500 + ?$$

$$154 = ?$$

Hence, the correct answer is option E.

31. Questions

Answer: B

$$83 + 14 = 97$$

$$97 + 28 = 125$$

$$125 + 42 = 167$$

$$167 + 56 = 223$$

$$223 + 70 = 293$$

Missing number: 293

Hence, the correct answer is option B.

32. Questions

Answer: A

$$1600 + 50 = 1650$$

$$1650 - 25 = 1625$$

$$1625 + 50 = 1675$$

$$1675 - 25 = 1650$$

$$1650 + 50 = 1700$$

Missing number: 1650

Hence, the correct answer is option A.

33. Questions

Answer: C

$$90 + 3^0 = 91$$

$$91 + 3^1 = 94$$

$$94 + 3^2 = 103$$

$$103 + 3^3 = 130$$

$$130 + 3^4 = 211$$

Missing number: 103

Hence, the correct answer is option C.

34. Questions

Answer: D

$$15 \quad 17 \quad 22 \quad 32 \quad 49 \quad 75$$

$$+2 \quad +5 \quad +10 \quad +17 \quad +26$$

$$+3 \quad +5 \quad +7 \quad +9$$

Missing number: 75

Hence, the correct answer is option D.

35. Questions

Answer: E

$$389 - 10^2 = 289$$

$$289 - 9^2 = 208$$

$$208 - 8^2 = 144$$

$$144 - 7^2 = 95$$

$$95 - 6^2 = 59$$

Missing number: 208

Hence, the correct answer is option E.

36. Questions

Answer: B

$$131 + 2^3 = 139$$

$$139 - 3^2 = 130$$

$$130 + 4^3 = 194$$

$$194 - 5^2 = 169$$

$$169 + 6^3 = 385$$

Wrong number: 203

Correct number: 385

Hence, the correct answer is option B.

37. Questions

Answer: A

$$43 \quad 49 \quad \mathbf{56} \quad 66 \quad 81 \quad 103$$

$$+6 \quad +7 \quad +10 \quad +15 \quad +22$$

$$+1 \quad +3 \quad +5 \quad +7$$

Wrong number: 55

Correct number: 56

Hence, the correct answer is option A.

38. Questions**Answer: E**

$$\begin{array}{cccccc} 119 & 128 & 146 & 182 & 254 & 398 \\ +9 & +18 & +36 & +72 & +144 & \\ *2 & *2 & *2 & *2 & & \end{array}$$

Wrong number: 120

Correct number: 119

Hence, the correct answer is option E.

39. Questions**Answer: D**

$$249 + 23 = 272$$

$$272 + 29 = 301$$

$$301 + 31 = 332$$

$$332 + 37 = 369$$

$$369 + 41 = 410$$

Wrong number: 411

Correct number: 410

Hence, the correct answer is option D.

40. Questions**Answer: C**

$$41 + 9^2 = 122$$

$$122 + 10^2 = 222$$

$$222 + 11^2 = 343$$

$$343 + 12^2 = 487$$

$$487 + 13^2 = 656$$

Wrong number: 485

Correct number: 487

Hence, the correct answer is option C.

41. Questions**Answer: B**

I). $x^2 - 23x + 132 = 0$

$$x^2 - 12x - 11x + 132 = 0$$

$$x(x - 12) - 11(x - 12) = 0$$

$$(x - 11)(x - 12) = 0$$

$$x = 11 \text{ or } 12$$

II). $y^2 - 19y + 88 = 0$

$$y^2 - 11y - 8y + 88 = 0$$

$$y(y - 11) - 8(y - 11) = 0$$

$$(y - 11)(y - 8) = 0$$

$$y = 11 \text{ or } 8$$

$$\text{So, } x \geq y$$

Hence, the correct answer is option B.

42. Questions

Answer: A

I). $x^2 - 17x + 72 = 0$

$$x^2 - 9x - 8x + 72 = 0$$

$$x(x - 9) - 8(x - 9) = 0$$

$$(x - 8)(x - 9) = 0$$

$$x = 8 \text{ or } 9$$

II). $y^2 - 9y + 18 = 0$

$$y^2 - 6y - 3y + 18 = 0$$

$$y(y - 6) - 3(y - 6) = 0$$

$$(y - 6)(y - 3) = 0$$

$$y = 6 \text{ or } 3$$

$$\text{So, } x > y$$

Hence, the correct answer is option A.

43. Questions

Answer: E

I). $x^2 - 27x + 170 = 0$

$$x^2 - 17x - 10x + 170 = 0$$

$$x(x - 17) - 10(x - 17) = 0$$

$$(x - 10)(x - 17) = 0$$

$$x = 10 \text{ or } 17$$

$$\text{II). } y^2 - 36y + 323 = 0$$

$$y^2 - 19y - 17y + 323 = 0$$

$$y(y - 19) - 17(y - 19) = 0$$

$$(y - 17)(y - 19) = 0$$

$$y = 17 \text{ or } 19$$

$$\text{So, } x \leq y$$

Hence, the correct answer is option E.

44. Questions

Answer: D

$$\text{I). } x^2 - 8x + 15 = 0$$

$$x^2 - 5x - 3x + 15 = 0$$

$$x(x - 5) - 3(x - 5) = 0$$

$$(x - 3)(x - 5) = 0$$

$$x = 3 \text{ or } 5$$

$$\text{II). } y^2 - 13y + 42 = 0$$

$$y^2 - 6y - 7y + 42 = 0$$

$$y(y - 6) - 7(y - 6) = 0$$

$$(y - 7)(y - 6) = 0$$

$$y = 7 \text{ or } 6$$

$$\text{So, } x < y$$

Hence, the correct answer is option D.

45. Questions

Answer: C

$$\text{I). } x^2 - 49 = 0$$

$$x^2 = 49$$

$$x = +7 \text{ or } -7$$

$$\text{II). } y^2 - 12y + 35 = 0$$

$$y^2 - 7y - 5y + 35 = 0$$

$$y(y - 7) - 5(y - 7) = 0$$

$$(y - 7)(y - 5) = 0$$

$$y = 7 \text{ or } 5$$

So, $x = y$ or relationship cannot be established.

Hence, the correct answer is option C.

@QuizByMickey